

Session 4 — Step-by-Step Setup Checklist

ExInt II: Research Designs in SME Research · WU Vienna · SS 2026 · May 6, 2026

Work through every step in order. Tick the box when done. Badges show where to work.

Example project — copy files or use as reference:

<https://github.com/vkiefner/sme-intl/tree/main>

Where to work:

Terminal

VS Code

Browser

GitHub

Zotero

A Install tools (skip steps you have already completed)

Browser

Step 1 — Create a GitHub account

Go to github.com → click Sign up → use your WU email address.

Confirm your email address when GitHub sends the verification link.

■ Skip if you already have a GitHub account.

Browser

Terminal

Step 2 — Install Python 3.10+

Go to python.org/downloads → download the latest Python 3.x installer.

■ Windows: on the first installer screen tick 'Add Python to PATH' before clicking Install.

Verify in a new terminal window:

```
python --version
```

■ Need 3.10+. If 'python not found' on Windows, try: `python3 --version`

Browser

VS Code

Step 3 — Install VS Code

Go to code.visualstudio.com → download and install.

Open VS Code → Extensions panel (Ctrl+Shift+X) → search and install:

```
Python (by Microsoft)
```

```
Pylance (by Microsoft)
```

■ VS Code detects your `.venv` folder automatically once it is created.

Browser

Terminal

Step 4 — Install Git

Go to git-scm.com/downloads → download and install.

Verify, then configure your identity (used in every commit):

```
git --version
```

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@wu.ac.at"
```

Browser

Step 5 — Install Quarto

Go to quarto.org → Download → install for your OS. Verify:

```
quarto --version
```

■ Quarto renders `research_note.md` to PDF. Used in Session 8 — install now.

Terminal

Step 6 — Install Taskfile

```
pip install go-task-bin
```

Verify:

```
task --version
```

B Create project folder and Python environment

Terminal

Step 7 — Create the folder structure

Navigate to where you keep your projects, then run:

```
mkdir sme-internationalization
```

```
cd sme-internationalization
```

```
mkdir data\raw data\processed code output\tables output\figures  
references
```

■ *Mac / Linux:*

```
mkdir -p data/raw data/processed code output/tables  
output/figures references
```

VS Code

Step 8 — Open the folder in VS Code

File → Open Folder → select sme-internationalization.

■ All file editing from here on happens inside VS Code.

Terminal

Step 9 — Create the virtual environment

```
python -m venv .venv
```

■ .venv is created inside the project. VS Code detects it automatically.

Terminal

Step 10 — Activate the virtual environment

```
.venv\Scripts\activate
```

■ *Mac / Linux:*

```
source .venv/bin/activate
```

■ Your prompt now shows (.venv). All pip installs go into this isolated environment.

Terminal

Step 11 — Install packages

```
pip install pandas numpy statsmodels linearmodels matplotlib  
seaborn wrds python-dotenv pyarrow jupyterlab ruff
```

■ Takes 1-2 minutes. Only needed once per project.

Terminal

Step 12 — Freeze package versions

```
pip freeze > requirements.txt
```

■ Anyone cloning your repo can reproduce your exact environment with: `pip install -r requirements.txt`

C Create and edit your project files

VS Code

Step 13 — Create .env — WRDS credentials

Create a new file in the project root called .env and add:

✎ **.env**

```
WRDS_USERNAME=your_wrds_username_here
```

■ .env is in .gitignore — it will NEVER be pushed to GitHub. See .env.example for reference.

VS Code

Step 14 — Edit README.md

Open README.md and fill in your name, research question, and at least one hypothesis:

📄 README.md

```
## Research Question
```

```
[Your draft research question]
```

```
## Hypotheses
```

```
- H1: [Your hypothesis]
```

■ A rough draft is fine — you will refine this in later sessions.

VS Code

Step 15 — Create pyproject.toml

Create a new file pyproject.toml in the project root:

📄 pyproject.toml

```
[project]
```

```
name = "sme-internationalization"
```

```
requires-python = ">=3.10"
```

```
[tool.ruff]
```

```
line-length = 88
```

```
exclude = [".venv"]
```

```
[tool.ruff.lint]
```

```
select = ["E4", "E7", "E9", "F"]
```

VS Code

Step 16 — Create Taskfile.yml

Create a new file Taskfile.yml in the project root:

📄 Taskfile.yml

```
version: '3'
```

```
tasks:
```

```
pull:
```

```
cmds:
```

```
- python code/01_pull_data.py
```

```
clean:
```

```
deps: [pull]
```

```
cmds:
```

```
- python code/02_clean.py
```

```
all:
```

```
deps: [clean]
```

■ Run task all in the terminal to execute all scripts in the correct order.

Browser

VS Code

Step 17 — Copy code scripts from the example project

Open the example project in your browser:

```
https://github.com/vkiefner/sme-intl/tree/main
```

Copy these four files into your code/ folder:

📄 code/01_pull_data.py | code/02_clean.py | code/03_descriptives.py | code/04_regression.py

■ You will adapt these scripts with your own research question in later sessions.

D Set up Zotero reference manager

Browser

Zotero

Step 18 — Install Zotero and browser connector

Go to zotero.org → Download Zotero (free) → install.

Also install the browser connector for Chrome or Firefox from zotero.org/download.

■ *The connector adds a button to your browser — one click saves any paper to Zotero.*

Browser

Zotero

Step 19 — Install Better BibTeX plugin

Go to retorque.re/zotero-better-bibtex → download the .xpi file.

In Zotero: Tools → Add-ons → drag the .xpi file into the window → restart Zotero.

Zotero

Step 20 — Configure auto-export to library.bib

In Zotero: right-click your library (left panel) → Export Library

Format: Better BibTeX → tick 'Keep updated' → save to:

📁 **references/library.bib**

■ *Every paper you add to Zotero now auto-updates library.bib. No manual step needed.*

Browser

Zotero

Step 21 — Add at least 3 papers

Find these online and click the browser connector button to import:

Johanson & Vahlne (1977) – The internationalization process of the firm ↵

Zaheer (1995) – Overcoming the liability of foreignness ↵

Lu & Beamish (2001) – The internationalization and performance of SMEs ↵

■ *After each import, check that references/library.bib updated in VS Code.*

E Initialize Git and push your project to GitHub

Browser

GitHub

Step 22 — Create a GitHub repository

Go to github.com/new and fill in:

Repository name: `sme-internationalization` ↵

Visibility: `Public` ↵

■ *Do NOT tick 'Add README', 'Add .gitignore', or 'Choose a license' — your project already has these.*

Click Create repository.

VS Code

Step 23 — Verify .gitignore

Open .gitignore and confirm these lines are present:

 **.gitignore**

.venv/	↵
data/raw/	↵
.env	↵
references/*.pdf	↵
__pycache__/	↵
*.pyc	↵
*.log	↵
*.tex	↵

■ This prevents credentials, raw data, and build files from being pushed to GitHub.

Terminal

Step 24 — Initialize Git, commit, and push

In your terminal (inside the project folder, venv active):

git init	↵
git branch -M main	↵
git add .	↵
git commit -m "initial project setup"	↵
git remote add origin https://github.com/YOUR_USERNAME/sme-internationalization	↵
git push -u origin main	↵

■ Replace YOUR_USERNAME with your GitHub username. Sign in via browser if prompted.

Browser

GitHub

Step 25 — Verify your repo on GitHub

Open in your browser: github.com/YOUR_USERNAME/sme-internationalization

You should see: README.md, requirements.txt, pyproject.toml, Taskfile.yml, code/, references/

■ If .env or data/raw/ appear in the repo, .gitignore is not working — ask for help.

Share your repo URL with the instructor.

F Final checklist — tick everything before leaving today

- ☐ GitHub account created and email confirmed
- ☐ Python 3.10+ installed (python --version works)
- ☐ VS Code installed with Python + Pylance extensions
- ☐ Git installed and configured with name + WU email
- ☐ Quarto installed (quarto --version works)
- ☐ Taskfile installed (task --version works)
- ☐ Project folder created with correct structure
- ☐ .venv created, activated, all packages installed
- ☐ requirements.txt frozen (pip freeze > requirements.txt)

- .env created with WRDS username (NOT committed to Git)
- README.md has your name, research question, and hypothesis
- pyproject.toml created with ruff config
- Taskfile.yml created
- Code scripts copied from example project into code/
- Zotero + Better BibTeX installed
- At least 3 papers in Zotero, library.bib auto-exporting
- .gitignore verified (.venv/, data/raw/, .env excluded)
- GitHub repo created, project pushed to main branch
- Repo URL shared with instructor

Next session (Session 5): We pull data from WRDS via the Python API. Make sure your WRDS credentials are in .env, your venv activates correctly, and your repo is on GitHub. If anything above is not ticked, ask the instructor before leaving today. Reference project: github.com/vkiefner/sme-intl